Lab: Specific Heat Capacity of Aluminum & Leakiness Test

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Purpose

To find the specific heat capacity of aluminum by experiment & compare the experimental value to reference material.

Introduction

This lab was designed to determine the specific heat capacity of aluminum by experiment using calorimetry. The specific heat capacity of a substance is the amount of energy required to raise the temperature of one gram of the substance by one degree Celsius. This experiment used a calorimeter, a device that minimizes heat and mass transfer between the system and its surroundings, to measure the heat transfer between aluminum and water. Additionally, the experiment included a leakiness test to determine the rate at which thermal energy leaks out of the calorimeter, which is important for accurate measurements.

The procedure involved heating aluminum pellets in a hot water bath until they reached the temperature of boiling water, $100°C$, then the aluminum was transferred to a calorimeter containing water with known mass and initial temperature. The maximum temperature reached by the water was recorded, as well as the time taken to reach that temperature. Finally, after waiting for 3 minutes, the temperature of the water once again was recorded to determine the rate of heat loss from the calorimeter.

The hot plate, boiling water, and aluminum pellets were handled with care to avoid burns and the rubber stopper was put on loosely to prevent pressure buildup. Additionally, the hot plate was not moved while in use and was unplugged after use to prevent accidents.

Observations and Results

Measurement	**Unit**	**Trial**
Mass of empty calorimeter	g	4.48
Mass of calorimeter + water	g	33.42
Mass of aluminum pellets (m_{Al})	g	15.09
Initial temp of water ($T_{i,w}$)	°C	25.9
Maximum temp reached (T_{max})	°C	32.0
Time to reach T_{max} (Δt)	s	102.49
Temp after 3-minute wait (T_{3min})	°C	31.8

Table 1: Experimental Data for Specific Heat of Aluminum

Analysis

Find the mass of water in the calorimeter

$m_{H_2O} = 33.42g - 4.48g = 28.94g$

Find the specific heat capacity of aluminum

$$Q = mc\Delta T$$

$$-m_{Al}c_{Al}(T_f - T_i) = m_w c_w (T_f - T_i)$$

$$c_{Al} = \frac{m_w c_w (T_f - T_i)}{-m_{Al}(T_f - T_i)}$$

$$c_{Al} = \frac{28.94g \cdot 4.184\frac{J}{g \cdot °C} \cdot (32.0°C - 25.9°C)}{-15.09g \cdot (32.0°C - 100°C)}$$

$$c_{Al} = 0.72\frac{J}{g \cdot °C}$$

∴ The specific heat capacity of aluminum is 0.72 J/(g dot °C)

Find the % error

$$\%error = \frac{|0.902\frac{J}{g \cdot °C} - 0.72\frac{J}{g \cdot °C}|}{0.902\frac{J}{g \cdot °C}} \cdot 100\%$$

$$\%error = 20\%$$

Leakiness Test

$$\text{rate} = \frac{32.0°C - 31.8°C}{180s}$$
$$= 0.001°\frac{C}{s}$$

$$\text{time correction} = 0.001°\frac{C}{s} \cdot 102.49s$$
$$= 0.1°C$$

$$\text{corrected } T_{\text{max}} = 32.0°C + 0.1°C$$
$$= 32.1°C$$

$$\% \text{ difference} = \frac{|32.1°C - 32.0°C|}{32.0°C} \cdot 100\%$$
$$= 0.3\%$$

∵ 0.3% < 5%

∴ the leakiness of the calorimeter does not significantly affect the results

Discussion

1. If more aluminum metal was used, a larger amount of heat would be transferred to the water, leading to a higher maximum temperature reached by the water. This would result in a larger ΔT for the aluminum, which could have lead to a more accurate calculation of the specific heat capacity, as the temperature change would larger and easier to measure.
2. There are many possible sources of experimental uncertainty. The most likely source is that the aluminum pellets were not at the assumed temperature of $100°C$. The aluminum may have been slightly cooler than boiling water, because the pellets were not directly in contact with the boiling water. Instead the pellets were in a small test tube, which may have caused the aluminum to heat up more slowly and not reach the full temperature of the boiling water. Additionally, heat loss to the surroundings could have occurred during the transfer process. This would lead to a lower calculated specific heat capacity, as the ΔT for the aluminum would be smaller than expected. Secondly, the calorimeter was not completely filled with water, as it had a significant amount of air in it. This would lead to a lower calculated specific heat capacity, as the air allows for water vapor to form, which is an endothermic process. Additionally, This would lead to a smaller ΔT for the water, which would decrease the calculated specific heat capacity of aluminum.

3. The system was not truly isolated and there were several sources of heat loss. The calorimeter was not perfectly insulated, and there were multiple place where mass and heat transfer could occur. The edge of the lid was uneven, and could not form a perfect seal, allowing for heat to escape. Additionally, the lid had a hole in it for the thermometer, which also allowed for heat to escape. Finally, the lid was not double insulated, like the container, so heat could have been lost through the lid as well. These sources of heat loss would have led to a lower calculated specific heat capacity, as the ΔT for the water would be smaller than expected.

Conclusion

In conclusion, the specific heat capacity of aluminum was experimentally determined to be $0.72 \frac{J}{g°C}$ with an error of 20% compared to the accepted value.